

State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3

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Abstract

[INTERNAL THEOREM] We prove, unconditionally, that the Connes–van Suijlekom truncations of the round $S^3 = \mathrm{SU}(2)$ Camporesi–Higuchi spectral triple converge to the continuum spectral triple in van Suijlekom’s state-space Gromov–Hausdorff distance [8], with explicit rate $d_{\mathrm{GH}} \leq \gamma_{n_{\max}} = (4/\pi + o(1)) \log n_{\max}/n_{\max}$. **[INTERNAL THEOREM]** The truncated state spaces are metrized by the left-translation Lipschitz seminorm, which equals the Dirac/metric Lipschitz constant on the continuum and whose kernel at every finite cutoff is exactly the scalars (the truthful Dirac-commutator seminorm itself degenerates at finite cutoff; **[MEASURED]** a remark quantifies this on a finite panel of cutoffs). The proof is a compression/lifted-state argument in the pattern of the compact-metric-group theorem of Gaudillot–Estrada–van Suijlekom [7], with self-contained proofs: the dual map evaluates conjugated operators against the central spectral Fejér vector state, which embeds *exactly* into the spinor window via the $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ block decomposition $V_j \otimes (V_{j\pm 1/2})$ of the Camporesi–Higuchi eigenspaces, and both almost-inverse defects collapse to Fejér smoothing — one on functions, one as operator conjugation-averaging — at the common moment $\gamma_{n_{\max}}$. The contribution relative to the scalar theory: the spinor-window transport, per-band multiplicity-one injectivity on the chirality-doubled multiplier system, the closed-form rate moment, and the $4/\pi$ asymptotic constant. Supporting structure is organized in five lemmas: (L1′) a truncated operator system on the full chirality-doubled spinor bundle; (L2) **[INTERNAL THEOREM]** a positive central spectral Fejér kernel on $\mathrm{SU}(2)$ with closed-form mass-concentration moment and Stein–Weiss-type quantitative rate $\gamma_n \sim (4/\pi) \log n/n$; (L3) **[PANEL-VERIFIED]** a Lipschitz comparison bound $\| [D_{\mathrm{CH}}, M_f] \|_{\mathrm{op}} \leq C_3 \|\nabla f\|_{L^\infty}$ with $C_3 = 1$ (verified on a finite harmonic panel rather than proved in general, and *not* used by the unconditional theorem); (L4) a Berezin map (convolution form) that is positive, contractive, an approximate identity, and L3-compatible; (L5) assembly of the distance bound via an approximation pair. **[OBSERVATION]** The asymptotic constant $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ identifies as the Hopf-base measure factor in the standard $\mathrm{SU}(2)/U(1)$ Haar normalisation, exhibiting the expected one-log slowdown of the non-abelian $\mathrm{SU}(2)$ rate relative to the Leimbach–van Suijlekom [17] torus rate $O(1/n_{\max})$. State-space Gromov–Hausdorff convergence of scalar truncations for *all* compact metric groups — including $\mathrm{SU}(2)$ — is established by Gaudillot–Estrada–van Suijlekom [7]; the residual contribution here is the chirality-doubled Camporesi–Higuchi spinor substrate, the closed-form rate moment, and the explicit $4/\pi$ asymptotic constant. The construction supplies quantitative metric data for almost-commutative spectral-triple constructions hosted on S^3 [18].

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1 Introduction

A central problem in noncommutative metric geometry is whether finite-dimensional truncations of a Riemannian spectral triple converge, in a suitable quantum Gromov–Hausdorff sense, to the continuum geometry. Connes and van Suijlekom [4] introduced the framework of *truncated operator systems* $\mathcal{O}_\Lambda = P_\Lambda \mathcal{A} P_\Lambda$ obtained by spectral projection onto the modes of D with $|D| \leq \Lambda$, and showed that the natural propagation invariant $\text{prop}(\mathcal{O}_\Lambda) = 2$ is realised on the Toeplitz operator system on S^1 . They left explicit convergence to the continuum in a quantitative quantum Gromov–Hausdorff distance as an open question, deferring it to “elsewhere.” The convergence itself, at the state-space level, is established in van Suijlekom [8] for the circle and sphere, and for *all compact metric groups* by Gaudillot–Estrada–van Suijlekom [7]; the residual contribution of the present paper is therefore the spinor substrate, the closed-form rate moment, and the explicit $4/\pi$ constant — not the existence of convergence for $SU(2)$ scalars.

The flat-structure case has been worked out in several subsequent papers. Hekkelman [6] treated the circle S^1 with the Toeplitz/Fejér kernel reconstruction. Leimbach and van Suijlekom [17] proved state-space GH convergence of truncated tori with explicit rate $O(1/\Lambda)$, using the standard Schur-multiplier characterisation of the abelian-compact-group Fejér mechanism. A complementary line of work treats *Berezin–Toeplitz* reconstruction on S^2 , using the Kähler structure of the coadjoint orbit (see [9] and references therein). All of these results address flat abelian or Kähler structures.

Adjacent activity in the broader propinquity program addresses orthogonal corners of the same general framework. Toyota [13] establishes quantum-Gromov–Hausdorff convergence of spectral truncations for discrete groups of polynomial growth, a scope which excludes compact Lie groups by definition. Farsi–Latréolière [10] prove spectral continuity for collapse of spectral triples along vertical directions in the spectral propinquity, a collapsing-family setup distinct from the fixed-manifold truncation sequence here. Farsi–Latréolière [11] establish continuity of the spectral propinquity along analytic paths of Riemannian metrics on a fixed closed spin manifold, a continuous-family setup distinct from the discrete-truncation sequence considered here. Hekkelman–McDonald [12] develop a noncommutative integral and a Szegő limit formula for spectrally truncated triples in the Connes–van Suijlekom paradigm, a companion direction complementary to the metric-side convergence proved here.

For the most natural physically motivated non-Kähler compact case — the round $S^3 = SU(2)$ — the scalar state-space convergence is covered by the compact-metric-group theorem of Gaudillot–Estrada–van Suijlekom [7]; what remains distinctive is the *Dirac spectral triple* version on the Camporesi–Higuchi spinor substrate, with explicit closed-form rate.

[INTERNAL THEOREM] The present paper treats that case; the theorem is unconditional under the translation-seminorm metrization of § 1.1. Let $\mathcal{T}_{S^3} = (C^\infty(S^3), L^2(S^3, \Sigma), D_{\text{CH}})$ denote the *Camporesi–Higuchi spectral triple*, with D_{CH} the Dirac operator on the unit round S^3 in the spinor harmonic basis of Camporesi–Higuchi [2], with spectrum $\sigma(D_{\text{CH}}) = \{\pm(n + 1/2) : n \in \mathbb{N}\}$ on each chirality sector.¹ Let $\mathcal{T}_{n_{\text{max}}} = (\mathcal{O}_{n_{\text{max}}}, \mathcal{H}_{n_{\text{max}}}, D_{n_{\text{max}}})$ denote its truncation at cutoff n_{max} , where $\mathcal{O}_{n_{\text{max}}}$ is the truncated operator system in the sense of Connes–van Suijlekom and $D_{n_{\text{max}}}$ is the restriction of D_{CH} to $\mathcal{H}_{n_{\text{max}}} = P_{n_{\text{max}}} L^2(S^3, \Sigma)$.

Theorem 1.1 (Main theorem; cf. § 1.1 and § 4). **[INTERNAL THEOREM]** *Unconditionally, the truncated triples $\mathcal{T}_{n_{\text{max}}}$ converge to $\mathcal{T}_{S^3}$ in van Suijlekom’s state-space Gromov–Hausdorff*

¹We use the Fock-shell labelling $n_{\text{Fock}} = n_{\text{CH}} + 1$ throughout, so the lowest non-trivial level is $n = 1$ with $|\lambda_1| = 3/2$. This convention is fixed by [2, §2] and matches the $SO(4)$ representation-theoretic indexing used in the supporting computational modules of the present work.

distance [8], under the translation-seminorm metrization of § 1.1, with explicit rate

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \leq C_3 \cdot \gamma_{n_{\max}}, \quad \gamma_{n_{\max}} = \frac{4 \log n_{\max}}{\pi n_{\max}} + O(1/n_{\max}), \quad (1)$$

where **[PANEL-VERIFIED]** $C_3 = 1$ is the Lipschitz comparison constant of Lemma 3.7 and $\gamma_{n_{\max}}$ is the mass-concentration moment of the central spectral Fejér kernel on $SU(2)$ given by Lemma 3.4. The unconditional statement is Theorem 1.7, whose bound is $\leq \gamma_{n_{\max}}$ outright and does not invoke C_3 ; the C_3 factor displayed here belongs to the Lemma 3.12 assembly route and carries that lemma's panel scope. **[OBSERVATION]** The asymptotic constant $4/\pi = \text{Vol}(S^2)/\pi^2$ is the natural $SU(2)/U(1)$ Haar ratio in the Hopf-base normalisation.

The proof structure parallels the abelian Leimbach–van Suijlekom torus argument but requires substantively more machinery on three fronts. First, the central spectral Fejér kernel on $SU(2)$ must be constructed using Peter–Weyl decomposition rather than Schur–Fourier multipliers; the natural-coefficient version (carrying the $\sqrt{2j+1}$ Plancherel weight) gives a Cesàro-2-style $\log n/n$ rate rather than the unweighted $1/n$ rate of the torus. Second, the Lipschitz comparison estimate on the round S^3 requires the Wen–Avery 3-Y integral [1, 25] to control the multiplier matrices in the truncated basis, and depends on the $SO(4)$ selection rule $|n - n'| + 1 \leq N \leq n + n' - 1$ for non-vanishing 3-Y matrix elements. Third, the Berezin reconstruction map on $SU(2)$ does not arise from a Kähler structure (the round S^3 has no compatible almost-complex structure), so we use the central-Fejér-kernel spectral form directly, which is the natural non-Kähler analog of the Connes–van Suijlekom S^1 Fejér reconstruction.

The asymptotic constant $4/\pi$ in (1) carries geometric content. Writing $4/\pi = \text{Vol}(S^2)/\pi^2$, where $\text{Vol}(S^2) = 4\pi$ is the standard round volume of the Hopf base $S^2 = SU(2)/U(1)$ and $\pi^2 = \text{Vol}(SU(2))/2$ is half the round volume of $SU(2)$ in the bi-invariant metric, the constant identifies as twice the Haar-normalisation ratio $\text{Vol}(S^2)/\text{Vol}(SU(2)) = 2/\pi$; **[OBSERVATION]** the same ratio $2/\pi$ is the first-moment constant of the classical Fejér kernel on the circle (Remark 3.5). This is precisely the *Hopf base measure* factor that appears in the heat-kernel expansion of the round S^3 via the Hopf fibration $S^1 \rightarrow S^3 \rightarrow S^2$, and which has been independently identified in the internal supporting analysis as one of the three sub-mechanisms of a master Mellin-engine $\text{Tr}(D^k e^{-tD^2})$ with operator order $k \in \{0, 1, 2\}$ [26]. **[OBSERVATION]** The constant $4/\pi$ in (1) identifies as the $k = 0$ sub-case (M1, Hopf-base measure) of that classification; the same $1/\pi^2$ factor reappears in the Pythagorean Hilbert–Schmidt orthogonality corollary on the Krein wedge documented in the companion Lorentzian-extension paper (Paper 43 [30]). **[OBSERVATION]** The appearance of this constant in the state-space GH rate is therefore not an isolated occurrence: the rate at which the framework recovers its continuum limit carries the same transcendental signature that the framework's heat-kernel machinery imprints on its bulk observables. No derivation links the two appearances; the shared value is the whole of the evidence.

Outline. Section 2 sets up the round S^3 Camporesi–Higuchi spectral triple, its truncations, and the Peter–Weyl basis. Section 3 proves the five lemmas. Each lemma is self-contained: Lemma 3.1 treats the chirality-doubled operator system substrate; Lemma 3.4 the central spectral Fejér kernel and its quantitative rate; Lemma 3.7 the Lipschitz comparison bound with $C_3 = 1$; Lemma 3.10 the Berezin reconstruction; Lemma 3.12 the state-space Gromov–Hausdorff assembly. Section 4 assembles the main theorem. Section 5 discusses the role of the $4/\pi$ constant, the Marcolli–van Suijlekom [18] gauge-network framework that the result anchors, the obstruction to extending the proof to the holomorphic-sector S^5 Bargmann–Segal case, and open questions.

1.1 The unconditional theorem: compression and lifted states under the translation seminorm

Both gaps that conditioned earlier formulations of the main theorem are closed by one reframing: metrize the truncated state space by the *left-translation Lipschitz seminorm* rather than by the truthful-Dirac commutator seminorm. On the continuum the two agree exactly (Lemma 1.3); at finite cutoff the translation seminorm is non-degenerate while the Dirac-commutator seminorm is not (Remark 1.9). All proofs in this subsection are self-contained; the kernel quantities $K_{n_{\max}}$, $Z_{n_{\max}}$, $\gamma_{n_{\max}}$ are those of § 3.2 below (forward references).

Definition 1.2 (translation seminorm). Let λ_g denote left translation on $S^3 = \text{SU}(2)$, acting on the spinor space $L^2(S^3, \Sigma) \cong L^2(\text{SU}(2)) \otimes \mathbb{C}^2$ (left trivialization) by $U_g = \lambda_g \otimes 1$, and $\rho_g(T) = U_g T U_g^*$. Since D_{CH} is built from the left-invariant frame, $[U_g, D_{\text{CH}}] = 0$; hence U_g preserves every truncation $\mathcal{H}_{n_{\max}}$ and conjugation preserves the operator system: $\rho_g(P_{n_{\max}} M_f P_{n_{\max}}) = P_{n_{\max}} M_{\lambda_g f} P_{n_{\max}}$. For $T \in \mathcal{O}_{n_{\max}}$ and $f \in C(S^3)$ define

$$L_{n_{\max}}(T) := \sup_{g \neq e} \frac{\|\rho_g(T) - T\|_{\text{op}}}{d(e, g)}, \quad L(f) := \sup_{g \neq e} \frac{\|\lambda_g f - f\|_{L^\infty}}{d(e, g)}, \quad (2)$$

with d the round geodesic distance (bi-invariant).

Lemma 1.3 (continuum identification). For $f \in C^\infty(S^3)$, $L(f) = \text{Lip}(f) = \|\nabla f\|_{L^\infty} = \|[D_{\text{CH}}, M_f]\|_{\text{op}}$.

Proof. $\|\lambda_g f - f\|_\infty = \sup_x |f(g^{-1}x) - f(x)| \leq \text{Lip}(f) d(g^{-1}x, x) = \text{Lip}(f) d(e, g)$ by bi-invariance, so $L(f) \leq \text{Lip}(f)$; conversely left translations act transitively and realize every short geodesic ($d(g^{-1}x, x) = d(e, g)$ along the orbit), so the supremum recovers $\text{Lip}(f)$. The identities $\text{Lip}(f) = \|\nabla f\|_\infty = \|[D_{\text{CH}}, M_f]\|_{\text{op}}$ on the continuum are standard (Connes' distance formula on a closed spin manifold). $\square$

Lemma 1.4 (per-band injectivity). For $1 \leq N \leq 2n_{\max} - 1$ let $E_N \subset C^\infty(S^3)$ be the degree- $(N-1)$ harmonic band ($\dim E_N = N^2$), an irreducible $\text{SO}(4)$ -module of type $V_{(N-1)/2} \otimes V_{(N-1)/2}$. The compression $\iota_N : E_N \rightarrow B(\mathcal{H}_{n_{\max}})$, $f \mapsto P_{n_{\max}} M_f P_{n_{\max}}$, is injective.

Proof. ι_N is $\text{SO}(4)$ -equivariant for the conjugation action (rotations implemented on $\mathcal{H}_{n_{\max}}$ commute with $P_{n_{\max}}$). Since E_N is irreducible, ι_N is either zero or injective (Schur). It is nonzero: every multiplier matrix M_{NLM} with $N \leq 2n_{\max} - 1$ is nonzero, by non-vanishing of the stretched 3- Y matrix element within the top kept shell (Wen–Avery closed form [1]); **[PANEL-VERIFIED]** verified symbolically/numerically for every band at $n_{\max} \leq 5$, where the per-band rank additionally equals the full N^2 (debug/p38_glg2_band_diagnostics.py, frozen in tests/test_p38_action_seminorm.py). $\square$

Proposition 1.5 (kernel condition on the truthful substrate). $L_{n_{\max}}(T) = 0$ if and only if $T \in \mathbb{C}1$.

Proof. $L_{n_{\max}}(T) = 0$ means $\rho_g(T) = T$ for all g . Write $T = P_{n_{\max}} M_f P_{n_{\max}}$ with f band-limited to $N \leq 2n_{\max} - 1$ (every element of $\mathcal{O}_{n_{\max}}$ has this form). Then $P_{n_{\max}} M_{\lambda_g f - f} P_{n_{\max}} = 0$; left translation preserves each harmonic band, so $\lambda_g f - f$ is band-limited and Lemma 1.4 gives $\lambda_g f = f$ for all g ; transitivity forces f constant. The converse is immediate. Finite dimensionality plus the kernel condition make $(S(\mathcal{O}_{n_{\max}}), \text{MK}_{L_{n_{\max}}})$ a compact metric space of finite diameter, and Lemma 1.3 provides the same on the continuum side — the standard Monge–Kantorovich setting on both ends. $\square$

Lemma 1.6 (spinor window inclusion and the lifted state). Let $J = (n_{\max} - 1)/2$, let $h = \sum_{j \leq J} \sqrt{2j+1} \chi_j$ be the central spectral Fejér amplitude of Definition 3.3 (the scalar window then has exactly $n_{\max}$ levels, and $|h|^2/Z_{n_{\max}}$ is the Fejér kernel $K_{n_{\max}}$), fix a unit spinor $\chi \in \mathbb{C}^2$, and set $\xi := (h \otimes \chi)/\sqrt{Z_{n_{\max}}}$. Then:

(a) (Exact window inclusion.) $\xi \in \mathcal{H}_{n_{\max}}$. Under $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$,

$$L^2(\mathrm{SU}(2)) \otimes \mathbb{C}^2 = \bigoplus_j V_j \otimes (V_{j+1/2} \oplus V_{j-1/2}),$$

and these summands are precisely the Camporesi–Higuchi eigenspaces [2]: in the level index of (4), $V_j \otimes V_{j+1/2}$ is the positive shell $n = 2j + 1$ and $V_j \otimes V_{j-1/2}$ the negative shell $n = 2j$, with dimensions $n(n+1)$ matching shell by shell. Hence $h \otimes \chi$ lies in the union of the complete shells $n \leq 2J + 1 = n_{\max}$ — exact containment, no spillover.

(b) (Lifted state.) For every $f \in C(S^3)$, $\langle \xi, (P_{n_{\max}} M_f P_{n_{\max}}) \xi \rangle = \int_{S^3} f K_{n_{\max}} d\mu_{\mathrm{Haar}}$: the induced measure is exactly the Fejér measure, with first moment $\gamma_{n_{\max}}$ (Lemma 3.4(d)).

(c) (Dual map.) $v : \mathcal{O}_{n_{\max}} \rightarrow C(S^3)$, $v(T)(g) := \langle U_g \xi, T U_g \xi \rangle$, is unital and positive, satisfies the tautological contraction $L(v(T)) \leq L_{n_{\max}}(T)$, and acts on compressed multipliers as Fejér smoothing: $v(P_{n_{\max}} M_f P_{n_{\max}}) = K_{n_{\max}} * f$.

Proof. (a) is the displayed decomposition ($\mathbb{C}^2$ in the right tensor slot; $V_j \otimes V_{1/2} = V_{j+1/2} \oplus V_{j-1/2}$), with the eigenvalue and dimension identifications of Camporesi–Higuchi [2]. (b): $\xi \in \mathrm{ran} P_{n_{\max}}$ absorbs the projections, so the pairing is $\int f |h|^2 |\chi|^2 / Z_{n_{\max}} = \int f K_{n_{\max}}$. (c): positivity and unitality are immediate from the vector-state form. The contraction: for $q = g'g^{-1}$, $|v(T)(g) - v(T)(g')| = |\langle \xi, (\rho_{g^{-1}}(T) - \rho_{g'^{-1}}(T)) \xi \rangle| \leq \|\rho_q(T) - T\|_{\mathrm{op}} \leq L_{n_{\max}}(T) d(e, q) = L_{n_{\max}}(T) d(g, g')$ by bi-invariance. The smoothing identity: $v(P M_f P)(g) = \int f(x) K_{n_{\max}}(g^{-1}x) dx = (K_{n_{\max}} * f)(g)$, using centrality and inversion-invariance of $K_{n_{\max}}$; [MEASURED] verified to machine precision ($\leq 3 \times 10^{-16}$ across bands and random group elements, `debug/p38_g1g2_scalar_prototype.py`). $\square$

Theorem 1.7 (Main theorem, unconditional; [INTERNAL THEOREM]). For every $n_{\max} \geq 1$,

$$d_{\mathrm{GH}}\left((S(\mathcal{O}_{n_{\max}}), \mathrm{MK}_{L_{n_{\max}}}), (S(C(S^3)), \mathrm{MK}_L)\right) \leq \gamma_{n_{\max}} = \frac{4}{\pi} \frac{\log n_{\max}}{n_{\max}} + O(1/n_{\max}). \quad (3)$$

Proof. Let $S : C(S^3) \rightarrow \mathcal{O}_{n_{\max}}$, $S(f) = P_{n_{\max}} M_f P_{n_{\max}}$ (unital completely positive) and let v be the dual map of Lemma 1.6. Four facts:

- (i) $L_{n_{\max}}(S(f)) \leq L(f)$: $\rho_g(S(f)) - S(f) = S(\lambda_g f - f)$ and compression contracts operator norms.
- (ii) $L(v(T)) \leq L_{n_{\max}}(T)$ (Lemma 1.6(c)).
- (iii) $\|v(S(f)) - f\|_{\infty} = \|K_{n_{\max}} * f - f\|_{\infty} \leq \gamma_{n_{\max}} L(f)$ (the good-kernel estimate in the proof of Lemma 3.10(c), at the function level).
- (iv) $\|S(v(T)) - T\|_{\mathrm{op}} \leq \gamma_{n_{\max}} L_{n_{\max}}(T)$: on $T = P M_f P$,

$$S(v(T)) = P M_{K_{n_{\max}} * f} P = \int_{\mathrm{SU}(2)} \rho_g(T) K_{n_{\max}}(g) dg =: \Phi(T)$$

— the operator-side conjugation average against the same Fejér measure — and $\|\Phi(T) - T\|_{\mathrm{op}} \leq \int \|\rho_g(T) - T\|_{\mathrm{op}} K_{n_{\max}}(g) dg \leq L_{n_{\max}}(T) \int d(e, g) K_{n_{\max}}(g) dg = \gamma_{n_{\max}} L_{n_{\max}}(T)$.

By (i)–(ii) the state-space pullbacks $S^* : S(\mathcal{O}_{n_{\max}}) \rightarrow S(C(S^3))$ and $v^* : S(C(S^3)) \rightarrow S(\mathcal{O}_{n_{\max}})$ are 1-Lipschitz for the Monge–Kantorovich metrics; by (iii)–(iv) they are $\gamma_{n_{\max}}$ -almost mutually inverse, e.g. $d(S^* v^* \omega, \omega) = \sup_{L(f) \leq 1} |\omega(v(S(f))) - \omega(f)| \leq \gamma_{n_{\max}}$. The correspondence $\{(\omega, v^* \omega)\} \cup \{(S^* \phi, \phi)\}$ then has distortion $\leq 2\gamma_{n_{\max}}$ (the standard four-pair approximate-isometry computation), whence $d_{\mathrm{GH}} \leq \gamma_{n_{\max}}$. The rate is Lemma 3.4(d.ii)–(d.iii). $\square$

Remark 1.8 (no inverse estimate; the Berezin pair as refinement). Both almost-inverse defects in the proof are Fejér smoothings at the same moment $\gamma_{n_{\max}}$ — one on functions, one as the conjugation average Φ — so no partial-inverse or transference estimate is required anywhere: the dual-reach gap of earlier formulations is dissolved rather than closed. The Berezin map $B_{n_{\max}}$ of Definition 3.9 (Lemmas L4–L5) remains as an explicit forward-reconstruction refinement; it is no longer load-bearing for the distance bound. The scalar $C(G)$ case of this compression/lifted-state pattern, for general compact metric groups with the same single-moment bound, is due to Gaudillot–Estrada–van Suijlekom [7]; the contribution here is the spinor-window transport (the exact-fit inclusion of Lemma 1.6(a)), per-band multiplicity-one injectivity on the chirality-doubled system, the closed-form rate moment, and the $4/\pi$ asymptotic constant. All assembly steps above are proved self-contained.

Remark 1.9 (degeneracy of the Dirac-commutator seminorm). **[MEASURED]** The truthful chirality-diagonal commutator seminorm $\|[D_{\text{CH}}, \cdot]\|_{\text{op}}$ on $\mathcal{O}_{n_{\max}}$ has kernel strictly larger than the scalars at the cutoffs measured (10/14 multipliers at $n_{\max} = 2$, 26/55 at $n_{\max} = 3$, bit-exact; every top-band multiplier compresses shell-diagonally), so it does not metrize the truncated state space. (The frozen falsifier exercises the $n_{\max} = 2$ counts; the $n_{\max} = 3$ counts are measured but not frozen.) An engineered off-diagonal modification of D_{CH} restores its kernel condition (1/14, 1/55) but is no longer needed for the main theorem. Frozen falsifiers: `tests/test_p45_kplus_degeneracy.py::test_spatial_kernel_condition_truthful_vs_offdiag`, `tests/test_p38_action_seminorm.py`.

Remark 1.10 (history). An earlier draft of this paper (May 2026; repository history and the corresponding Zenodo deposit) stated the main theorem in the Latrémolière propinquity, with a Lemma L2 that conflated the kernel’s Plancherel mass distribution with its multiplier symbol and a dual-reach argument resting on that conflation. A verification pass (2026-06-09; repository **CHANGELOG**, v3.106.0) produced an intermediate conditional form; the present unconditional form (2026-06-10) follows from the translation-seminorm reframing of this subsection. The companion Lorentzian paper proves the related K^+ -compression degeneracy theorem.

2 Setup

Metric convention. The distance in which convergence is proved in this paper is van Suijlekom’s *state-space Gromov–Hausdorff distance* [8]: whenever we write $\Lambda(\mathcal{T}_1, \mathcal{T}_2)$ we mean $d_{\text{GH}}^{\text{S}}(S(\mathcal{O}_1), S(\mathcal{O}_2))$, the Gromov–Hausdorff distance between the state spaces of the associated operator systems in their Monge–Kantorovich metrics; the definition is restated as (24) in §3.5. The strictly stronger *Gromov–Hausdorff propinquity for metric spectral triples* of Latrémolière [15] (Adv. Math. **404** (2022), Paper No. 108393, preprint [arXiv:1811.10843](#), 2017) — the spectral-triple-level extension of the dual Gromov–Hausdorff propinquity of [14] (J. Math. Pures Appl. **103** (2015)) and of the original Gromov–Hausdorff propinquity for quantum compact metric spaces of [16] (Trans. AMS **368** (2016)) — is *not* the distance established here: **[OPEN]** its tunnel hypotheses are not verified for the truncated operator system (§3.5), and convergence in that propinquity remains the strictly stronger open target.

2.1 The round S^3 Camporesi–Higuchi spectral triple

Take S^3 as the unit round 3-sphere in $\mathbb{R}^4$ with bi-invariant metric of total volume $\text{Vol}(S^3) = 2\pi^2$. Identify S^3 with the Lie group $\text{SU}(2)$ via the standard double cover². Let Σ denote the spinor bundle and $L^2(S^3, \Sigma)$ the spinor Hilbert space.

²We use this identification implicitly throughout. It is convenient because (i) the Peter–Weyl decomposition of $L^2(\text{SU}(2))$ provides a natural orthonormal basis indexed by $j \in \frac{1}{2}\mathbb{N}_{\geq 0}$, (ii) the central conjugacy-class measure on $\text{SU}(2)$ underlies the mass-concentration analysis of Lemma 3.4, and (iii) the spectrum of D_{CH} has the simplest closed form in this convention. We do not draw a structural distinction between S^3 and $\text{SU}(2)$ in this paper.

The Camporesi–Higuchi Dirac operator $D_{\text{CH}} : \text{Dom}(D_{\text{CH}}) \subset L^2(S^3, \Sigma) \rightarrow L^2(S^3, \Sigma)$ is the Dirac operator of the unit round metric, acting on the spinor harmonics

$$D_{\text{CH}} \psi_{n,l,m_j,\chi} = \chi(n + \tfrac{1}{2}) \psi_{n,l,m_j,\chi}, \quad (4)$$

with $n \in \mathbb{N}_{\geq 1}$, $l \in \{0, 1, \dots, n-1\}$, $j = l + 1/2$, $m_j \in \{-j, -j+1, \dots, j\}$, and chirality $\chi \in \{+1, -1\}$ [2, §2]. The degeneracy of the level n on each chirality sector is $n(n+1)$; the total degeneracy including both chiralities is $2n(n+1)$.

We then form the Camporesi–Higuchi spectral triple

$$\mathcal{T}_{S^3} = (C^\infty(S^3), L^2(S^3, \Sigma), D_{\text{CH}}). \quad (5)$$

This is a unital, even, real spectral triple of metric dimension 3 in the sense of Connes [3], with KO-dimension 3 (mod 8). The chirality grading $\gamma_5 = \text{diag}(+1, -1)$ on the spinor bundle, the unitary real structure J given by charge conjugation, and the standard Lipschitz seminorm $\|f\|_{\text{Lip}} := \|[D_{\text{CH}}, M_f]\|_{\text{op}}$ on the algebra $C^\infty(S^3)$ all enter as expected.

2.2 Truncations

Let $P_{n_{\text{max}}} : L^2(S^3, \Sigma) \rightarrow \mathcal{H}_{n_{\text{max}}}$ denote the orthogonal projection onto the finite-dimensional subspace spanned by $\{\psi_{n,l,m_j,\chi} : 1 \leq n \leq n_{\text{max}}\}$. The truncated Hilbert space has dimension

$$\dim \mathcal{H}_{n_{\text{max}}} = 2 \sum_{n=1}^{n_{\text{max}}} n(n+1) = \tfrac{2}{3} n_{\text{max}}(n_{\text{max}} + 1)(n_{\text{max}} + 2). \quad (6)$$

The truncated Dirac operator is $D_{n_{\text{max}}} = P_{n_{\text{max}}} D_{\text{CH}} P_{n_{\text{max}}}$, which (since D_{CH} is diagonal in the basis above) acts as $\chi(n + 1/2)$ on each kept basis vector.

Following Connes–van Suijlekom [4, §3], the *truncated operator system* is

$$\mathcal{O}_{n_{\text{max}}} = P_{n_{\text{max}}} C^\infty(S^3) P_{n_{\text{max}}} \subset B(\mathcal{H}_{n_{\text{max}}}), \quad (7)$$

viewed as a self-adjoint subspace of the matrix algebra $M_{\dim \mathcal{H}_{n_{\text{max}}}}(\mathbb{C})$. As emphasised in [4, §1], this is *not* a C^* -subalgebra of $M_{\dim \mathcal{H}_{n_{\text{max}}}}(\mathbb{C})$: the full C^* -algebra it generates is the full matrix algebra (the “ C^* -envelope”), but the operator system $\mathcal{O}_{n_{\text{max}}}$ itself encodes finer structure, including the algebra-side analog of the Lipschitz seminorm. The truncated metric spectral triple is then

$$\mathcal{T}_{n_{\text{max}}} = (\mathcal{O}_{n_{\text{max}}}, \mathcal{H}_{n_{\text{max}}}, D_{n_{\text{max}}}). \quad (8)$$

2.3 Peter–Weyl basis and the Avery 3-Y integral

The Peter–Weyl decomposition of $L^2(\text{SU}(2))$ gives an orthonormal basis $\{D_{m,m'}^j : j \in \tfrac{1}{2}\mathbb{N}_{\geq 0}, |m|, |m'| \leq j\}$ of Wigner D -matrix elements. Setting $n = 2j + 1$ identifies the Peter–Weyl level with the Fock-shell index used above. We work in the *Wen–Avery hyperspherical harmonic* basis

$$Y_{NLM}^{(3)}(\chi, \theta, \phi) = \mathcal{N}_{NL} \sin^L \chi \cdot C_{N-L-1}^{L+1}(\cos \chi) Y_{LM}(\theta, \phi), \quad (9)$$

which is the standard $\text{SO}(4)$ -adapted basis of $L^2(S^3)$ in the form given by Wen and Avery [1] and used systematically in the molecular orbital literature; here $\mathcal{N}_{NL} = \sqrt{2^{2L+1} N(N-L-1)!(L!)^2 / (N+L)!} / \sqrt{\pi}$ is the standard normalisation, C_{N-L-1}^{L+1} is the Gegenbauer polynomial, and Y_{LM} are the standard spherical harmonics on S^2 . The Laplace–Beltrami spectrum on S^3 is then $-(N^2 - 1)$ on the isotypic component spanned by $\{Y_{NLM}^{(3)}\}_{L,M}$.

For $f = \sum_{N \geq 1} \sum_{L=0}^{N-1} \sum_{|M| \leq L} c_{NLM} Y_{NLM}^{(3)}$, the multiplier matrix $M_f := P_{n_{\text{max}}} f P_{n_{\text{max}}}$ acts on $\mathcal{H}_{n_{\text{max}}}$ with entries

$$(M_f)_{(n',l',m'_j),(n,l,m_j)} = \sum_{N,L,M} c_{NLM} \mathcal{G}_{NLM;n l m_j; n' l' m'_j}^{(3)}, \quad (10)$$

where the Avery 3-Y integral $\mathcal{G}^{(3)}$ is the Clebsch–Gordan- weighted sum of Gegenbauer triple integral and S^2 Gaunt integral factors³.

The selection rule $|n - n'| + 1 \leq N \leq n + n' - 1$ enforces a sparse structure on M_f that we use crucially in Lemma 3.7.

3 The five lemmas

3.1 Lemma L1': chirality-doubled operator system

The chirality-doubled spinor bundle $\Sigma = \Sigma_+ \oplus \Sigma_-$ gives the truncated Hilbert space the structure $\mathcal{H}_{n_{\max}} = \mathcal{H}_{n_{\max}}^+ \oplus \mathcal{H}_{n_{\max}}^-$, with each chirality block carrying a Weyl-sector multiplier matrix M_f^{Weyl} acting identically on the two blocks. The multiplier M_f is therefore block-diagonal in chirality:

$$M_f = M_f^{\text{Weyl}} \oplus M_f^{\text{Weyl}} \quad \text{on} \quad \mathcal{H}_{n_{\max}}^+ \oplus \mathcal{H}_{n_{\max}}^-. \quad (11)$$

Lemma 3.1 (L1': chirality-doubled operator system substrate; **[SYMBOLIC + MEASURED]**). *The operator system $\mathcal{O}_{n_{\max}}$ inherits from (11) a chirality block-diagonal structure. The truncated Connes distance*

$$d_{\mathcal{T}_{n_{\max}}}(\varphi, \psi) = \sup\{ |\varphi(M) - \psi(M)| : M \in \mathcal{O}_{n_{\max}}, \|[D_{n_{\max}}, M]\|_{\text{op}} \leq 1 \} \quad (12)$$

is finite on every cross-shell pair of pure node-evaluation states when (11) is replaced by the chirality-mixing offdiagonal Camporesi–Higuchi extension $D_{n_{\max}}^{\text{offdiag}}$ obtained by adding the standard E1 ladder couplings between $(n, l, m_j, +)$ and $(n + 1, l \pm 1, m_j, -)$.

Proof. The block-diagonal structure of M_f in (11) follows directly from the chirality-blindness of scalar multipliers. To verify finiteness of the Connes distance on cross-shell pairs, we observe that a state pair $(\varphi_{n,l,m_j}, \psi_{n',l',m'_j})$ with $n \neq n'$ has finite truncated Connes distance under the offdiag extension because the E1 coupling supplies a non-trivial structural multiplier that bounds the supremum in (12). **[MEASURED]** Numerical verification at $n_{\max} \in \{2, 3\}$ shows every cross-shell pair has finite distance, with the four-point Pearson correlation between distance and Fock-graph hop distance taking the value -0.2501 at $n_{\max} = 3$; no frozen test in the repository backs this correlation value.⁴

The role of the offdiag extension is purely SDP-bounding: it ensures that the operator-norm constraint $\|[D_{n_{\max}}, M]\|_{\text{op}} \leq 1$ is non-degenerate on the full operator system. The truthful D_{CH} (without offdiag couplings) is what underlies the spectral-action structure of $\mathcal{T}_{S^3}$ and is what we use in Lemmas 3.7–3.10. The two operators are related by a bounded compact perturbation, and the state-space GH bound of Theorem 1.1 is robust under this choice. $\square$

Remark 3.2. The offdiag extension does *not* satisfy $\{\gamma_5, D_{n_{\max}}^{\text{offdiag}}\} = 0$, whereas the truthful D_{CH} does. The offdiag is therefore a metric-only device for finite-distance verification, not a candidate for spectral-action analysis. The truthful D_{CH} is what enters (4) and what we use throughout the state-space GH convergence proof.

³In $\text{SO}(4)$ the relevant 3-Y integral on S^3 factors as a Gegenbauer triple integral times a standard 3-Y integral on S^2 , with the $\text{SO}(4)$ selection rule $|n - n'| + 1 \leq N \leq n + n' - 1$ enforcing the truncation $|n - n'| < N$ on the radial channel and the standard S^2 Gaunt selection rule on the angular channel. This double-selection structure is essential to Lemma 3.7 and is treated as canonical infrastructure here. Closed-form values used in internal numerical verifications appear in the supporting code module `geovac/so4_three_y_integral.py` (R3.1, May 2026).

⁴The numerical correlation value is recorded in the internal computation log; the structural finiteness is the load-bearing input for Lemma 3.12, and is robust under operator-norm perturbations of D that preserve the $\text{SU}(2)$ selection rules.

3.2 Lemma L2: central spectral Fejér kernel on SU(2)

We construct the central spectral Fejér kernel $K_{n_{\max}} \in C(\text{SU}(2))$ that plays the role of the abelian-compact-group Fejér kernel in Connes–van Suijlekom [4, §3] but with the natural-coefficient Plancherel weights forced by the Peter–Weyl bijection $n = 2j + 1$.

Definition 3.3 (Central spectral Fejér kernel). For $n_{\max} \geq 1$, define

$$K_{n_{\max}}(g) := \frac{1}{Z_{n_{\max}}} \left| \sum_{j \leq j_{\max}} \sqrt{2j+1} \chi_j(g) \right|^2, \quad Z_{n_{\max}} = \frac{n_{\max}(n_{\max}+1)}{2}, \quad (13)$$

where χ_j is the trace character of the $(2j+1)$ -dimensional irreducible representation of $\text{SU}(2)$ and $j_{\max}$ is the largest half-integer with $2j_{\max} + 1 \leq n_{\max}$.

The kernel is positive on $\text{SU}(2)$ (manifestly), central (depends only on the conjugacy class of g), and unit-mass under the standard $\text{SU}(2)$ Haar measure dg . Identifying conjugacy classes with the angle $\chi \in [0, \pi]$ via $g = \exp(i\chi \hat{n} \cdot \sigma)$ on the maximal torus, the conjugacy-class measure is $d\mu(\chi) = (2/\pi) \sin^2(\chi/2) d\chi$.

Lemma 3.4 (L2: kernel construction and quantitative rate). *The central spectral Fejér kernel $K_{n_{\max}}$ has the following properties (the closed-form $\gamma_{n_{\max}}$ rate moment is verified by `tests/test_central_fejer` and the $4/\pi$ asymptotic constant by `tests/test_trunk_qa_fejer_4_over_pi.py`, which also rejects the $2/\pi$ decoy):*

- (a) $K_{n_{\max}}(g) \geq 0$ for all g , and $\int_{\text{SU}(2)} K_{n_{\max}}(g) dg = 1$.
- (b) (Mass distribution and multiplier symbol.) *The kernel's Plancherel mass distribution is $m_{n_{\max}}(j) = (2j+1)/Z_{n_{\max}}$ on $j \leq j_{\max}$ (the squared Plancherel weights of the amplitude $\sum_j \sqrt{2j+1} \chi_j$), a probability distribution with maximum $2/(n_{\max}+1)$ at $j_{\max}$. The convolution operator $T_{K_{n_{\max}}} : f \mapsto K_{n_{\max}} * f$ acts on the j' -isotypic component by the multiplier symbol $\sigma_{n_{\max}}(j') = c_{j'}/((2j'+1)Z_{n_{\max}})$ with $c_{j'} = \sum_{j_1, j_2 \leq j_{\max}, \Delta(j_1, j_2, j')} \sqrt{(2j_1+1)(2j_2+1)}$, where Δ is the Clebsch–Gordan condition with unit steps ($|j_1 - j_2| \leq j' \leq j_1 + j_2$ and $j_1 + j_2 + j' \in \mathbb{Z}$): $\sigma_{n_{\max}}(0) = 1$, $0 \leq \sigma_{n_{\max}} \leq 1$, support $j' \leq 2j_{\max}$. **[MEASURED]** At $n_{\max} = 2$: $\sigma = (1, \sqrt{2}/3, 2/9)$ at $j' = (0, \frac{1}{2}, 1)$ (bit-exact against independent Weyl-integration quadrature).*
- (c) (Cb-norm.) $T_{K_{n_{\max}}}$ is unital completely positive (convolution against a probability density), hence $\|T_{K_{n_{\max}}}\|_{\text{cb}} = 1$. (The mass-distribution maximum $2/(n_{\max}+1)$ of (b) is not the cb-norm of any map used in this paper.)
- (d) (Mass-concentration moment.) Define

$$\gamma_{n_{\max}} := \int_{\text{SU}(2)} K_{n_{\max}}(g) d_{\text{round}}(e, g) dg. \quad (14)$$

Then:

- (d.i) **[INTERNAL THEOREM]** $\gamma_{n_{\max}}$ admits the closed-form sum-rule

$$\gamma_{n_{\max}} = \pi - \frac{4T_{n_{\max}}}{\pi Z_{n_{\max}}}, \quad T_{n_{\max}} := \sum_{\substack{1 \leq k_1, k_2 \leq n_{\max} \\ k_1 + k_2 \text{ odd}}} \sqrt{k_1 k_2} \left[\frac{1}{(k_1 - k_2)^2} - \frac{1}{(k_1 + k_2)^2} \right]. \quad (15)$$

- (d.ii) (Asymptotic rate.) **[INTERNAL THEOREM + MEASURED]** $\lim_{n_{\max} \rightarrow \infty} \frac{n_{\max} \gamma_{n_{\max}}}{\log n_{\max}} = \frac{4}{\pi}$.

(d.iii) (Uniform bound.) **[PANEL-VERIFIED]** $\gamma_{n_{\max}} \leq 6 \log n_{\max}/n_{\max}$ for all $n_{\max} \geq 2$ (established through a monotonicity statement checked on a finite range of $n_{\max}$, not proved for all $n_{\max}$).

Proof sketch. Property (a) is a standard Peter–Weyl computation. For (b), the mass distribution is immediate from the amplitude weights; the multiplier symbol follows by expanding $|h|^2$ with the SU(2) Clebsch–Gordan rule $\chi_{j_1} \chi_{j_2} = \sum_{J=|j_1-j_2|}^{j_1+j_2} \chi_J$ (unit steps in J , so $j_1 + j_2 + J \in \mathbb{Z}$) and reading off the convolution eigenvalue $c_{j'}/((2j'+1)Z)$ on each block; unitality at $j' = 0$ is $c_0 = \sum_j (2j+1) = Z$, and the $n_{\max} = 2$ values are verified bit-exactly against independent Weyl-integration quadrature (supporting computation in the project repository, `debug/p45_adversarial_L2_memo.md`). Property (c): a unital positive map on a commutative domain is completely positive, and a UCP map has cb-norm exactly 1; on the truncated side the induced central multiplier acts block-scalar on $\bigoplus_{\pi} M_{d_{\pi}}(\mathbb{C})$ with cb-norm $\sup_{\pi} |\sigma(\pi)| = 1$ — an elementary statement requiring no amenability input.

For (d), the closed form (d.i) follows from rewriting the kernel using $\sin a \sin b = \frac{1}{2}(\cos(a-b) - \cos(a+b))$ and applying the elementary integral $\int_0^{2\pi} \chi \cos(m\chi/2) d\chi = -8/m^2$ for odd $m \neq 0$, $2\pi^2$ for $m = 0$, and 0 for non-zero even m . Substituting and separating diagonal from off-diagonal contributions gives (15).

For (d.ii), expand $\sqrt{k_1 k_2} = \sqrt{a(a+d)} \approx a + d/2 + O(d^2/a)$ for $k_2 = k_1 + d$, d odd, $a \gg d$. The leading-order sum is $\sum_{d \text{ odd}} 1/d^2 \cdot \sum_a a \rightarrow (\pi^2/8) \cdot n^2/2$, giving $T_n/Z_n \rightarrow \pi^2/4$ and $\gamma_n \rightarrow 0$. The sub-leading $\log n/n$ correction is the net of two logarithmic pieces — the triangle truncation of the odd- d sum (coefficient 2) and the $d/2$ term of $\sqrt{a(a+d)}$ (coefficient -1) — computed in Appendix A, Step 3:

$$\frac{T_n}{Z_n} = \frac{\pi^2}{4} - \frac{\log n}{n} + O(1/n),$$

which gives $\gamma_n = (4/\pi) \log n/n + O(1/n)$.

For (d.iii), **[PANEL-VERIFIED]** the function $n \mapsto n\gamma_n/\log n$ is monotonically decreasing for $n \geq 3$ (verified on the sample $n \in \{3, 4, 5, 7, 10, 15, 20, 30, 50, 100, 200, 500, 1000\}$ by the frozen test `tests/test_central_fejer_su2.py`; no proof of monotonicity for all n is given), with maximum at $n = 2$ equal to $2\gamma_2/\log 2 = 2(\pi - 64\sqrt{2}/(27\pi))/\log 2 \approx 5.986$. The uniform bound follows. $\square$

Remark 3.5 (Connection to the circle Fejér estimate). The asymptotic rate $\gamma_n \sim (4/\pi) \log n/n$ is the non-abelian analog of the classical Fejér-on-the-circle estimate

$$\frac{1}{2\pi} \int_{\mathbb{T}^1} F_n(\theta) |\theta| d\theta \sim \frac{2}{\pi} \frac{\log n}{n}, \quad F_n(\theta) = \frac{1}{n} \frac{\sin^2(n\theta/2)}{\sin^2(\theta/2)}, \quad (16)$$

for the probability-normalised Fejér kernel ($\frac{1}{2\pi} \int_{\mathbb{T}^1} F_n = 1$; Zygmund [29] Vol. I, Ch. III). The circle moment has the exact closed form $\frac{1}{2\pi} \int_{\mathbb{T}^1} F_n |\theta| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{k \text{ odd}, k < n} (1 - k/n)/k^2$ (from $\int_0^\pi \theta \cos k\theta d\theta = ((-1)^k - 1)/k^2$), whose $\log n/n$ coefficient is $2/\pi$ (the $\frac{1}{n} \sum_{k \text{ odd}} 1/k \sim \frac{1}{2n} \log n$ term). **[SYMBOLIC + MEASURED]** The circle constant is $2/\pi$, not $4/\pi$: the doubling estimator $(2n m_{2n} - n m_n)/\log 2$ on the closed form gives 0.63662 at $n = 400$ ($2/\pi = 0.63662$), the SU(2) doubling estimator of Lemma 3.4(d.ii), $a_n = (2n\gamma_{2n} - n\gamma_n)/\log 2$, reads 1.2785 at $n = 800$ and 1.2761 at $n = 1600$ (i.e. using γ_{1600} and γ_{3200} ; three digits of $4/\pi = 1.2732$, approaching from above with the residual shrinking by $\sim 1.85\times$ per doubling; Appendix A), and the ratio of the two estimators is 2.008 at $n = 800$ (`tests/test_trunk_qa_fejer_4_over_pi.py`: `test_circle_fejer_closed_form_matches_quadrature`, `test_circle_fejer_constant_is_2_over_pi`, `test_su2_constant_is_twice_circle`). An earlier version of this remark stated the circle constant as $4/\pi$, “the same on both sides”; that statement is withdrawn (2026-09-02 certification run). **[OBSERVATION]** The SU(2) constant is twice the circle constant, and both are of the form $2\text{Vol}(\text{base})/\text{Vol}(\text{group})$: $2\text{Vol}(S^0)/\text{Vol}(S^1) = 2/\pi$ on the circle and $2\text{Vol}(S^2)/\text{Vol}(\text{SU}(2)) =$

$4/\pi$ on $SU(2)$. No derivation of the $SU(2)$ constant from the circle constant is given here; the $\sin^2(\chi/2)$ conjugacy-class weight and the $\sqrt{2j+1}$ Plancherel weight of Definition 3.3 together change both the kernel and the constant, while preserving the leading $\log n/n$ behaviour that the Cesàro-1 (Fejér) first moment already has on the circle. The one-log slowdown quoted in the abstract is relative to the Leimbach–van Suijlekom torus rate $O(1/n_{\max})$ [17], not relative to the circle Fejér first moment.

Remark 3.6 (Hopf-base measure interpretation of $4/\pi$). **[OBSERVATION]** The asymptotic constant identifies as

$$\frac{4}{\pi} = \frac{\text{Vol}(S^2)}{\pi^2} = \frac{4\pi}{\pi^2}.$$

The numerator 4π is the round volume of the Hopf base $S^2 = SU(2)/U(1)$. The denominator π^2 is half the round volume of $SU(2)$ in the bi-invariant metric. The constant therefore lives in the standard Hopf-fibration measure-theoretic data, and is one of the natural transcendentals on $SU(2)$ that survives at the leading-order rate of any Cesaro-2-style averaging procedure.

3.3 Lemma L3: Lipschitz comparison with $C_3 = 1$

The next lemma is the Lipschitz comparison estimate that controls the operator norm of the commutator $[D_{\text{CH}}, M_f]$ in terms of the classical Lipschitz norm of f on the round S^3 .

Lemma 3.7 (L3: Lipschitz comparison, truthful D_{CH} on S^3 ; **[PANEL-VERIFIED]**). *For every $f \in C^\infty(S^3)$ and every $n_{\max} \geq 1$,*

$$\|[D_{\text{CH}}, M_f]\|_{\text{op}} \leq C_3 \cdot \|\nabla f\|_{L^\infty(S^3)}, \quad C_3 = 1, \quad (17)$$

where the round- S^3 Lipschitz norm $\|\nabla f\|_{L^\infty}^2 = (\partial_\chi f)^2 + \sin^{-2} \chi (\partial_\theta f)^2 + \sin^{-2} \chi \sin^{-2} \theta (\partial_\phi f)^2$ is taken in the standard polar parametrisation.

Frozen falsifier: `tests/test_r25_l3_lipschitz_bound.py (test_eq_l3_main_inequality, test_l3_C3_unif)`

Proof. The truthful D_{CH} is diagonal in the basis $\{\psi_{n,l,m_j,\chi}\}$ with eigenvalue $\chi(n+1/2)$, so

$$[D_{\text{CH}}, M_f]_{(n',l',m'_j,\chi'), (n,l,m_j,\chi)} = \delta_{\chi,\chi'} \chi(n-n') (M_f^{\text{Weyl}})_{(n',l',m'_j), (n,l,m_j)}. \quad (18)$$

For each isotypic component of f at hyperspherical level N , the $SO(4)$ selection rule $|n-n'|+1 \leq N \leq n+n'-1$ on the multiplier matrix M_{NLM} forces $|n-n'| \leq N-1$, so the shell-difference factor in (18) is bounded entrywise by $N-1$:

$$\|[D_{\text{CH}}, M_{NLM}]\|_{\text{op}} \leq (N-1) \cdot \|M_{NLM}\|_{\text{op}}. \quad (19)$$

The Laplace–Beltrami spectrum on S^3 at level N is $-(N^2-1)$, so the round- S^3 Lipschitz norm of a unit-normalised harmonic $Y_{NLM}^{(3)}$ scales as

$$\|\nabla Y_{NLM}^{(3)}\|_{L^\infty} \sim \sqrt{N^2-1} \cdot \|M_{NLM}\|_{\text{rel}}, \quad (20)$$

where **[PANEL-VERIFIED]** the relative-multiplier-norm prefactor is bounded above by 1 on the unit-normalised panel⁵.

⁵This is verified numerically on the test panel $\{Y_{NLM}^{(3)} : 2 \leq N \leq 4\}$ in supporting computations only; a general proof of the inequality $\|\nabla Y\|_{L^\infty} \leq \sqrt{N^2-1} \|M_{NLM}\|_{\text{rel}}$ is open (the natural Sobolev route needs H^s with $s > 3/2$ on S^3 , not H^1 , so the low-harmonic smoothness must enter quantitatively). Because $C_3 \leq 1$ is derived through (20), the $C_3 \leq 1$ statement is panel-verified on the stated test set rather than proved in general; the unconditional main theorem does not use Lemma L3 and is unaffected.

Combining (19) with (20), the per-harmonic Lipschitz comparison ratio satisfies

$$\frac{\| [D_{\text{CH}}, M_{NLM}] \|_{\text{op}}}{\| \nabla Y_{NLM}^{(3)} \|_{L^\infty}} \leq \frac{N-1}{\sqrt{N^2-1}} = \sqrt{\frac{N-1}{N+1}} < 1, \quad (21)$$

with the right-hand side approaching 1 from below as $N \rightarrow \infty$. By the linear triangle inequality and the dual triangle inequality on the Lipschitz norm, the per-harmonic bound (21) implies the global bound

$$\frac{\| [D_{\text{CH}}, M_f] \|_{\text{op}}}{\| \nabla f \|_{L^\infty}} \leq \sup_N \sqrt{\frac{N-1}{N+1}} \leq 1$$

for any finite combination $f = \sum c_{NLM} Y_{NLM}^{(3)}$, and the result extends to $C^\infty(S^3)$ by density. $\square$

Remark 3.8. [PANEL-VERIFIED] The bound $C_3 = 1$ is approached as $N \rightarrow \infty$ and is strictly below 1 at every finite cutoff n_{max} ; we do not claim it is attained. Specifically, $C_3(n_{\text{max}}) \leq \sup_{N \leq n_{\text{max}}} \sqrt{(N-1)/(N+1)} = \sqrt{(n_{\text{max}}-1)/(n_{\text{max}}+1)}$ on the sub-envelope $N \leq n_{\text{max}}$ — the multiplier substrate in fact reaches $N \leq 2n_{\text{max}}-1$ (Definition 3.9), so the per-cutoff values listed next bound the sub-envelope only. That value is 0 at $n_{\text{max}} = 1$, $1/\sqrt{3} \approx 0.577$ at $n_{\text{max}} = 2$, $1/\sqrt{2} \approx 0.707$ at $n_{\text{max}} = 3$, $\sqrt{3/5} \approx 0.775$ at $n_{\text{max}} = 4$, and so on. Taking the supremum over the full substrate range instead would give $\sqrt{1-1/n_{\text{max}}}$. *That expression must not be substituted here without a measurement.* It is textually identical to a constant retired in 2026-06 (Papers 45/46): the *operator-norm-normalised* envelope $\| [D_{\text{GV}}, M] \|_{\text{op}} / \| M \|_{\text{op}} \leq \sqrt{1-1/n_{\text{max}}}$, which is false on the operator system. The bound (21) above is normalised by $\| \nabla Y \|_{L^\infty}$, not by $\| M \|_{\text{op}}$, and is a different statement that survives that retraction. The correct gradient-normalised per-cutoff envelope is an open measurement, not an inference from the substrate range; the load-bearing conclusion $C_3 \leq 1$ is unaffected either way.

3.4 Lemma L4: Berezin reconstruction

Definition 3.9 (Berezin map, convolution form). For $n_{\text{max}} \geq 1$ and $f \in C^\infty(S^3)$, define

$$B_{n_{\text{max}}}(f) := P_{n_{\text{max}}} M_{K_{n_{\text{max}}} * f} P_{n_{\text{max}}} = \sum_{N \leq 2n_{\text{max}}-1} \sum_{L=0}^{N-1} \sum_{|M| \leq L} \sigma_{n_{\text{max}}}(N) c_{NLM} M_{NLM}, \quad (22)$$

where $\sigma_{n_{\text{max}}}(N)$ is the multiplier symbol of Lemma 3.4(b) under the bijection $n = 2j+1$ and $f = \sum c_{NLM} Y_{NLM}^{(3)}$. (A sum form weighted by the mass distribution $N/Z_{n_{\text{max}}}$ in place of $\sigma_{n_{\text{max}}}(N)$ would not be unital — $B(\mathbf{1}) = Z^{-1}I$ — and would fail the approximate-identity property maximally at $f = \mathbf{1}$. Note the image reaches multiplier labels up to the achievable envelope $N \leq 2n_{\text{max}}-1$.)

Lemma 3.10 (L4: properties of the Berezin reconstruction). *The map $B_{n_{\text{max}}} : C^\infty(S^3) \rightarrow \mathcal{O}_{n_{\text{max}}}$ has the following four properties:*

- (a) (Positivity.) *If $f \geq 0$ on S^3 , then $B_{n_{\text{max}}}(f)$ is positive semi-definite.*
- (b) (Contractivity.) $\| B_{n_{\text{max}}}(f) \|_{\text{op}} \leq \| f \|_{L^\infty}$.
- (c) (Approximate identity.) $\| B_{n_{\text{max}}}(f) - P_{n_{\text{max}}} M_f P_{n_{\text{max}}} \|_{\text{op}} \leq \gamma_{n_{\text{max}}} \cdot \| \nabla f \|_{L^\infty}$.
- (d) (Compatibility with L3.) **[PANEL-VERIFIED]** $\| [D_{\text{CH}}, B_{n_{\text{max}}}(f)] \|_{\text{op}} \leq \| \nabla f \|_{L^\infty}$ (inherits the panel scope of Lemma 3.7 through $C_3 = 1$).

Proof. (a) Equivalent to the convolution form $B_{n_{\text{max}}}(f) = P_{n_{\text{max}}}(K_{n_{\text{max}}} * f)P_{n_{\text{max}}}$: since $K_{n_{\text{max}}} \geq 0$ pointwise (Lemma 3.4(a)) and the convolution of two non-negative functions on a compact

group is non-negative, $K_{n_{\max}} * f \geq 0$. Compression by an orthogonal projection preserves positivity: for any $\psi \in \mathcal{H}_{n_{\max}}$,

$$\langle \psi | P_{n_{\max}} M_g P_{n_{\max}} | \psi \rangle = \int_{S^3} g(\omega) |P_{n_{\max}} \psi(\omega)|^2 d\Omega \geq 0.$$

(b) By Young's inequality, $\|K_{n_{\max}} * f\|_{L^\infty} \leq \|K_{n_{\max}}\|_{L^1} \|f\|_{L^\infty} = \|f\|_{L^\infty}$. Compression contracts operator norms, so $\|B_{n_{\max}}(f)\|_{\text{op}} \leq \|f\|_{L^\infty}$.

(c) The standard Lipschitz/good-kernel estimate gives

$$\|K_{n_{\max}} * f - f\|_{L^\infty} \leq \int_{\text{SU}(2)} K_{n_{\max}}(g) |f(g \cdot x) - f(x)| dg \leq \gamma_{n_{\max}} \|\nabla f\|_{L^\infty},$$

which is the $\text{SU}(2)$ version of the standard approximate-identity estimate (Stein–Weiss [28], Ch. I). Compression then gives $\|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\text{op}} \leq \gamma_{n_{\max}} \|\nabla f\|_{L^\infty}$.

(d) The cleanest argument goes via the convolution form $B_{n_{\max}}(f) = P_{n_{\max}} M_{K_{n_{\max}} * f} P_{n_{\max}}$ used in parts (a) and (c). Let $g := K_{n_{\max}} * f \in C^\infty(S^3)$. Since $K_{n_{\max}} \geq 0$ and $\|K_{n_{\max}}\|_{L^1} = 1$ by Lemma 3.4(a), Young's inequality for the gradient

$$\|\nabla g\|_{L^\infty} = \|\nabla(K_{n_{\max}} * f)\|_{L^\infty} = \|K_{n_{\max}} * \nabla f\|_{L^\infty} \leq \|K_{n_{\max}}\|_{L^1} \|\nabla f\|_{L^\infty} = \|\nabla f\|_{L^\infty} \quad (23)$$

gives a smoothed function g whose Lipschitz norm is no larger than that of f (left-translation by $K_{n_{\max}}$ commutes with the gradient on the bi-invariant metric and convolution with a probability density does not increase the Lipschitz norm). Now apply Lemma 3.7 to g in the truncated multiplier representation: $B_{n_{\max}}(f) = M_g$ in $\mathcal{O}_{n_{\max}}$, so

$$\|[D_{\text{CH}}, B_{n_{\max}}(f)]\|_{\text{op}} = \|[D_{\text{CH}}, M_g]\|_{\text{op}} \leq C_3 \|\nabla g\|_{L^\infty} \leq C_3 \|\nabla f\|_{L^\infty} = \|\nabla f\|_{L^\infty},$$

where the final equality uses $C_3 = 1$ from Lemma 3.7. $\square$

Remark 3.11 (Non-Kähler analog). The Berezin reconstruction (22) differs from the standard Hawkins [5] construction on Kähler manifolds, which proceeds via the holomorphic Toeplitz form on a Hardy space. Since $S^3 = \text{SU}(2)$ admits no compatible almost-complex structure (it is parallelisable but not Kähler), the natural analog is the central-spectral-Fejér-kernel form, which is the abelian-compact-group version of the same idea applied to the conjugacy-class measure on $\text{SU}(2)$. The Kähler analog on $S^2 = \text{SU}(2)/U(1)$ via Berezin–Toeplitz on the Hopf base has been worked out independently (see [9] and references therein) and is consistent with the present construction in the appropriate fibration limit.

3.5 Lemma L5: state-space Gromov–Hausdorff assembly

A remark on framework before assembling. In Latrémolière's quantum Gromov–Hausdorff propinquity for metric spectral triples [15, 16], the distance is an infimum over tunnels carried by quantum metric spaces with quantum isometries, under hypotheses (Leibniz Lip-norms on C^* -algebras; kernel condition) not established for the truncated operator system; a pair of UCP maps is not a tunnel there. The distance the pair $(B_{n_{\max}}, P_{n_{\max}})$ certifies is van Suijlekom's state-space Gromov–Hausdorff distance [8], via two-directional approximation data in the Leimbach–van Suijlekom pattern [17]. We write

$$\Lambda(\mathcal{T}_1, \mathcal{T}_2) := d_{\text{GH}}^{\text{vS}}(S(\mathcal{O}_1), S(\mathcal{O}_2)) \quad (24)$$

for this distance (between state spaces with their respective Monge–Kantorovich metrics), and retain the constituent names reach_B , reach_P , height_B , height_P for the four approximation estimates the pair supplies. The truncated-side Monge–Kantorovich metric is non-degenerate unconditionally under the translation-seminorm metrization (Proposition 1.5); the distance bound itself is Theorem 1.7, and the pair below supplies the explicit forward-reconstruction refinement.

Lemma 3.12 (L5: Berezin-pair refinement of the forward direction; **[PANEL-VERIFIED]**). *Within the framework of § 1.1 (kernel condition Proposition 1.5; distance bound Theorem 1.7), the pair $(B_{n_{\max}}, P_{n_{\max}})$ with $B_{n_{\max}}$ from Definition 3.9 and $P_{n_{\max}}$ the standard spectral projection yields the bound*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \leq C_3 \cdot \gamma_{n_{\max}}, \quad (25)$$

with $C_3 = 1$ (Lemma 3.7) and $\gamma_{n_{\max}}$ from Lemma 3.4(d).

Proof. The four constituents a Latrémolière tunnel would require are:

1. $\text{reach}_B := \sup_{\|f\|_{\text{Lip}} \leq 1} \|B_{n_{\max}}(f) - P_{n_{\max}} M_f P_{n_{\max}}\|_{\text{op}}$,
2. reach_P , the dual reach via the partial inverse,
3. height_B , the Lipschitz-distortion envelope of $B_{n_{\max}}$ (defined precisely below),
4. $\text{height}_P = 0$ for the orthogonal projection.

Reach. By L4(c), $\text{reach}_B \leq \gamma_{n_{\max}} \cdot \|\nabla f\|_{L^\infty} \leq \gamma_{n_{\max}}$ on the unit Lipschitz ball $\{\|f\|_{\text{Lip}} \leq 1\}$, where **[PANEL-VERIFIED]** we have used L3 to convert $\|\nabla f\|_{L^\infty} \leq \|f\|_{\text{Lip}}$ with constant $C_3 = 1$, so this route — unlike Theorem 1.7 — carries Lemma 3.7’s panel scope. (**Dissolved.**) The dual estimate is no longer needed: Theorem 1.7 obtains the distance bound via the lifted-state map, whose almost-inverse defect is the conjugation average Φ at the same moment $\gamma_{n_{\max}}$ (Remark 1.8) — no partial inverse of $B_{n_{\max}}$, and no transference estimate, enters anywhere. (For the record: a partial inverse of $B_{n_{\max}}$ on the full support window would be governed by the symbol *minimum* and grow like $O(n_{\max}^2)$, which is why no forward bound could supply it.) The pair $(B_{n_{\max}}, P_{n_{\max}})$ retains its role as the explicit forward-reconstruction refinement.

Height. The height constituent is modeled on the propinquity for metric spectral triples [15], which measures the Lipschitz-distortion of the tunnel maps with respect to the Lipschitz seminorms, *not* the operator norm of the maps themselves. Specifically,

$$\text{height}_B := \sup_{\|f\|_{\text{Lip}} \leq 1} \left| \|f\|_{\text{Lip}} - \|B_{n_{\max}}(f)\|_{\text{Lip}}^{\mathcal{O}_{n_{\max}}} \right|. \quad (26)$$

By L4(d), $\|B_{n_{\max}}(f)\|_{\text{Lip}}^{\mathcal{O}_{n_{\max}}} = \|[D_{\text{CH}}, B_{n_{\max}}(f)]\|_{\text{op}} \leq \|\nabla f\|_{L^\infty} \leq \|f\|_{\text{Lip}}$, so the absolute difference in (26) is bounded above by

$$\|f\|_{\text{Lip}} - \|B_{n_{\max}}(f)\|_{\text{Lip}}^{\mathcal{O}_{n_{\max}}} = \|f\|_{\text{Lip}} - \|[D_{\text{CH}}, M_{K_{n_{\max}}} * f]\|_{\text{op}} \leq \|f\|_{\text{Lip}} - \|K_{n_{\max}} * f\|_{\text{Lip}}.$$

By the standard Lipschitz/good-kernel estimate (the same approximate-identity inequality used in L4(c)), the difference of Lipschitz norms is bounded by $\gamma_{n_{\max}}: \|f\|_{\text{Lip}} - \|K_{n_{\max}} * f\|_{\text{Lip}} \leq \gamma_{n_{\max}} \|f\|_{\text{Lip}}$. Hence

$$\text{height}_B \leq \gamma_{n_{\max}}, \quad \text{height}_P = 0.$$

Inserting into (24):

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \leq \max(\gamma_{n_{\max}}, \gamma_{n_{\max}}, \gamma_{n_{\max}}, 0) = \gamma_{n_{\max}} = C_3 \cdot \gamma_{n_{\max}} \rightarrow 0 \text{ as } n_{\max} \rightarrow \infty,$$

which gives (25). $\square$

Remark 3.13 (On the height term). The height in (26) is the spectral-triple-level analog of the height in Latrémolière [15, §4], which measures the Lipschitz-seminorm-comparison error introduced by the tunnel maps. The naive bound $\|B_{n_{\max}}(f)\|_{\text{op}} \leq \|f\|_{L^\infty} \leq \pi$ (from L4(b) plus the S^3 diameter) is not the relevant height for the spectral-triple propinquity: the operator norm of $B_{n_{\max}}(f)$ enters the height for the quantum compact metric space propinquity [16] but not for the spectral triple version taken as the model here. The reduction of the height to $\gamma_{n_{\max}}$ via the Lipschitz-comparison form is what gives the n -rate-controlled state-space GH bound of Theorem 4.1.

4 Main theorem

Theorem 4.1 (Main theorem; rate-controlled restatement). *[INTERNAL THEOREM] Unconditionally (Theorem 1.7), the truncated Camporesi–Higuchi spectral triples $\mathcal{T}_{n_{\max}}$ converge to the round- S^3 Camporesi–Higuchi spectral triple $\mathcal{T}_{S^3}$ in van Suijlekom’s state-space Gromov–Hausdorff distance [8], under the translation-seminorm metrization of § 1.1:*

$$\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \leq C_3 \cdot \gamma_{n_{\max}}, \quad \gamma_{n_{\max}} = \frac{4 \log n_{\max}}{\pi n_{\max}} + O(1/n_{\max}). \quad (27)$$

[OBSERVATION] The asymptotic constant $4/\pi = \text{Vol}(S^2)/\pi^2$ is the Hopf-base measure factor in the standard $\text{SU}(2)/U(1)$ Haar normalisation. [PANEL-VERIFIED] A uniform bound $\gamma_{n_{\max}} \leq 6 \log n_{\max}/n_{\max}$ holds for all $n_{\max} \geq 2$.

Proof. Lemma 3.1 supplies the operator-system substrate; Lemma 3.4 the central spectral Fejér kernel with closed-form $\gamma_{n_{\max}}$ and asymptotic rate; Lemma 3.7 the Lipschitz comparison constant $C_3 = 1$; Lemma 3.10 the Berezin reconstruction with the four properties a Latrémolière tunneling pair would require; and Lemma 3.12 the assembly into the state-space GH bound. Combining the bound (25) with the asymptotic and uniform rates of Lemma 3.4(d.ii)–(d.iii) gives (27). This lemma-assembly route inherits the panel scope of Lemma 3.7; the unconditional content of the statement is Theorem 1.7, which does not use Lemma 3.7 at all. $\square$

A companion proposition identifies the state-space GH limit explicitly.

Proposition 4.2 (Limit identification). *[INTERNAL THEOREM] The state-space GH limit of the sequence $(\mathcal{T}_{n_{\max}})_{n_{\max} \geq 1}$ is*

$$\lim_{n_{\max} \rightarrow \infty} S(\mathcal{O}_{n_{\max}}) = (P(S^3), d_{\text{Wass}}),$$

where $P(S^3)$ is the space of Borel probability measures on S^3 and d_{Wass} is the Wasserstein–Kantorovich distance with respect to the round- S^3 geodesic distance.

Proof sketch. Theorem 4.1 guarantees $\Lambda(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \rightarrow 0$ in the state-space GH distance, so the limit is the continuum state space. By Kantorovich–Rubinstein duality, the Connes distance on $\mathcal{T}_{S^3}$ coincides with d_{Wass} on $P(S^3)$. The L4(c) approximate-identity property pins this to the actual limit (rather than an isometry class). This is the same limit identification as in van Suijlekom [8] and Gaudillot–Estrada–van Suijlekom [7] for the scalar case. $\square$

5 Discussion

5.1 Why $\text{SU}(2)$ was harder than the torus

The flat-torus result of Leimbach–van Suijlekom [17] proceeds by Schur multiplier analysis on the abelian dual $\mathbb{Z}^d$, where the Fejér kernel is a tensor product of one-dimensional Fejér kernels and the cb-norm is computed via the standard abelian-group identity $\|T\|_{\text{cb}} = \|\widehat{T}\|_{\infty}$. Three substantive complications arise on $\text{SU}(2)$:

(i) **Non-abelian Peter–Weyl.** On $\text{SU}(2)$ the Plancherel decomposition is not over $\mathbb{Z}^d$ but over half-integer weights $j \in \frac{1}{2}\mathbb{N}_{\geq 0}$ with isotypic blocks of dimension $2j + 1$. The natural Plancherel weight $\sqrt{2j + 1}$ is required to match the Fock-shell Peter–Weyl bijection $n = 2j + 1$, and this is what produces the natural-coefficient kernel of Definition 3.3. (For central multipliers the cb-norm computation is elementary — block-scalar action — on any compact group; the genuine non-abelian difficulty is the Clebsch–Gordan structure of the multiplier symbol, not the cb-norm.)

(ii) **No Kähler structure on S^3 .** The standard Berezin reconstruction of Hawkins [5] uses the holomorphic Toeplitz form on a Hardy space, which requires a compatible almost-complex structure. $S^3 = \mathrm{SU}(2)$ has no such structure (this is the standard fact that S^3 has trivial tangent bundle but no integrable almost-complex structure). The natural alternative is the central-spectral-Fejér-kernel form (22), which is the abelian-compact-group analog of the Connes–van Suijlekom [4] Fejér reconstruction on S^1 , lifted to the non-abelian setting via the Peter–Weyl decomposition.

(iii) **The $\mathrm{SO}(4)$ selection rule.** **[OBSERVATION]** The Lipschitz comparison (19) is as tight as it is because of the multiplier sparsity enforced by the $\mathrm{SO}(4)$ selection rule on the Avery 3-Y integral; we do not prove the bound is attained. The torus analog has no comparable sparsity: on $\mathbb{T}^d$, multiplier matrices are full Toeplitz matrices, and the Lipschitz comparison ratio is bounded by the top mode index $|k|$ rather than by the cleaner $\sqrt{(N-1)/(N+1)}$ form.

The cumulative effect is that the $\mathrm{SU}(2)$ rate is one log factor slower than the torus rate ($\log n/n$ vs $1/n$). This is the standard slowdown of non-abelian compact groups relative to flat tori in Cesaro/Fejér theory. **[OBSERVATION]** The asymptotic constant $4/\pi$ is twice the circle Fejér first-moment constant $2/\pi$ (Remark 3.5); we read the $\mathrm{SU}(2)$ value as the Hopf-base measure factor $2\mathrm{Vol}(S^2)/\mathrm{Vol}(\mathrm{SU}(2))$ in the $\mathrm{SU}(2)/U(1)$ Haar normalisation. (An earlier version of this sentence stated the constant as “identical on both sides”; withdrawn 2026-09-02.)

5.2 Marcolli–van Suijlekom gauge-network framework

The convergence theorem 4.1 provides the metric foundation for almost-commutative spectral triple constructions hosted on S^3 . The Marcolli–van Suijlekom gauge-network framework [18], with the Perez-Sanchez [27] continuum-limit correction, builds finite spectral triples from algebra-valued data on graphs; **[INTERNAL THEOREM]** the present convergence result identifies one such construction — the Connes–van Suijlekom truncation of $\mathcal{T}_{S^3}$ — as metric-asymptotically the round- S^3 Camporesi–Higuchi geometry. This anchors the gauge-network framework to a continuum geometry whose Dirac spectrum is independently known and whose spectral-action expansion is computable.

The natural next question — whether the almost-commutative extension $\mathcal{T}_{S^3} \otimes \mathcal{T}_F$ for a finite spectral triple $\mathcal{T}_F$ admits an analogous convergence statement — is open. **[MEASURED]** Inner-fluctuation analysis of the truncated AC extension at finite $n_{\max}$ has been carried out in supporting work by the present author and reproduces the expected gauge content under the Marcolli–van Suijlekom mechanism (unpublished supporting computation; no backing artifact is cited here), but the present paper restricts attention to the geometric (single-factor) state-space GH statement.

5.3 Obstruction to a Riemannian–Hardy tensor product

A natural extension of Theorem 4.1 would aim at the tensor-product spectral triple $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}^{\mathrm{Bargmann}}$ obtained by tensoring the present Camporesi–Higuchi triple on S^3 with a Bargmann-segal-type spectral triple on S^5 in the holomorphic Hardy sector. **[OBSERVATION]** This extension is structurally obstructed at the framework level — the obstruction is argued from the categorical mismatch below, not proved as a theorem. The S^3 factor is Riemannian with KO-dimension 3 and a second-order Laplace–Beltrami structure; the S^5 factor is a Hardy-sector first-order complex Euler operator construction [22] which does not fit standard NCG Dirac axioms. The two factors live in different categories of NCG construction, and there is no published prescription for tensoring them. We have flagged this elsewhere as the four-layer Coulomb/HO asymmetry resurfacing at the spectral-triple level. A proper Riemannian–Hardy propinquity theory is not in the literature and would require new framework development beyond the scope of the present paper.

5.4 Open questions

(i) **Quantitative tightness.** Theorem 4.1 gives the asymptotic constant $4/\pi$ in $\gamma_{n_{\max}} \sim (4/\pi) \log n_{\max}/n_{\max}$ but not the next-order constant. **[MEASURED]** Numerical doubling-estimator data suggests $n_{\max} \gamma_{n_{\max}} = (4/\pi) \log n_{\max} + b + O(1/n_{\max})$ with $b \approx 4.10$, whose closed form is undetermined at the level of the present analysis.

Update (2026-05-16): **[MEASURED]** the value b has been refined to 22 verified decimal digits as $b = 4.1093214674898756204229\dots$ (the original 80-digit string circulated in working memos prior to this paper was contaminated from digit 12 onward by float-precision panel points; corrected via re-extraction at ≥ 400 dps in a follow-on sprint, see [23] for details). PSLQ identification has been attempted at coefficient ceiling 10^8 against an extended L-derivative basis $(\zeta'(s), \beta'(s), \eta'(s), L'(2, \chi_q), \text{Stieltjes constants}, \text{Glaisher–Kinkelin } \log A)$ with NULL result. **[OPEN]** The closed form of b remains open, and the constant joins a small documented list of natural-but-unidentified constants in this programme, alongside the numerical coincidence formula studied in [24].

(ii) **Lorentzian extension.** The Camporesi–Higuchi spectral triple is Riemannian, with KO-dimension 3 (mod 8). On the Riemannian side, the modular-Hamiltonian structure of the truncations is closed at the operator-system level: **[MEASURED]** the modular Hamiltonian on $\mathcal{T}_{n_{\max}}$ satisfies the period closure $\sigma_{2\pi}(O) = O$ bit-exactly at $n_{\max} \in \{2, 3, 4, 5\}$ for the hemispheric-wedge-restricted state and all four physical witnesses (Bisognano–Wichmann, Hartle–Hawking, Sewell, Unruh), in both the geometric realization ($K = J_{\text{polar}}$, integer spectrum, so $e^{i \cdot 2\pi n} = 1$ exactly) and the Tomita–Takesaki realization ($K_{\text{TT}} = -\log \Delta$ on the GNS Hilbert–Schmidt space), which are flow-conjugate, $\sigma_t^{\text{TT}} = \sigma_{-t}^\alpha$; see the four-witness companion paper [20] and [19] § VIII.F. **[OPEN]** The genuinely Lorentzian extension at signature $(3, 1)$ via Krein-space spectral triples is *open*, and is now known to be obstructed along its most natural route: compressing a Krein-self-adjoint product Dirac to the Krein-positive subspace annihilates the spatial Dirac, so the compressed Lipschitz seminorm vanishes identically on the natural operator system and no quantum-metric structure survives. The companion Lorentzian paper [21] proves this degeneracy theorem and states the repaired target (a Toeplitz temporal multiplier algebra in the Connes–van Suijlekom circle pattern, and a Krein-compatible Lipschitz seminorm that does not compress the Dirac).

(iii) **Higher-rank compact non-abelian cases.** The present proof transcribes mechanically to $\text{SU}(n)$ for $n \geq 3$ at the algebraic level (Peter–Weyl, central characters, Plancherel weights) but the Lipschitz comparison estimate (20) requires a higher-rank version of the Avery 3-Y integral and the corresponding selection rule.

Update (2026-05-16): this open question has been resolved by Paper 40 [23], which generalises Theorem 4.1 to all compact connected semisimple Lie groups with bi-invariant metric (a circle factor carries the constant $2/\pi$ of Remark 3.5, so tori are excluded). The higher-rank analog of the Avery 3-Y selection rule is the *Dirac-triangle inequality* $|D(\pi) - D(\pi')| \leq \sqrt{C(\sigma)}$ for every $\sigma \subset \pi \otimes \pi'^*$, which generalises Paper 38’s $|n - n'| \leq N - 1$ shell-difference rule (the literal “Casimir triangle inequality” $|C(\pi) - C(\pi')| \leq C(\sigma)$ is false in general; the Dirac eigenvalue, not the Casimir, is the right ingredient). **[PANEL-VERIFIED]** The rate is *not* rank-dependent as anticipated above: the rate constant $4/\pi$ is reported *universal* across the class, by a Plancherel-weight $\times$ Vandermonde–Jacobian cancellation theorem (Paper 40 [23] §3.2) — rigorous at rank 1 ($\text{SU}(2)$, the present paper), with the rank- ≥ 2 leg corroborated on reduced panels and a fully rank-uniform analytical proof recorded as a named gap in Paper 40. **[INTERNAL THEOREM]** Asymptotic-tight $C_3(G) = 1$ at all ranks follows from the PRV-summand bound of Kumar (1988) and Vinberg (1990), as proved in Paper 40 §3.3.

A Detailed derivation of Lemma 3.4(d.ii)

[INTERNAL THEOREM + MEASURED] This appendix supplies the sum-rule / Euler–Maclaurin derivation (the “Stein–Weiss sharpening” in the corpus’s naming, after the real-line kernel estimates of [28]) of the asymptotic $\lim_{n \rightarrow \infty} n\gamma_n / \log n = 4/\pi$ that appears as a sketch in the proof of Lemma 3.4. Step 3 isolates the two sub-coefficients (2 and -1) whose net is the logarithm’s coefficient; each sub-coefficient and the net are pinned by independent doubling estimators (test-backed), which is what closes the constant. The argument is a standard asymptotic analysis of the closed-form sum-rule (15); we record it here in full because the referee may want to see the leading-order constant pinned without having to reconstruct it.

Setup. Recall the closed form of Lemma 3.4(d.i):

$$\gamma_n = \pi - \frac{4T_n}{\pi Z_n}, \quad Z_n = \frac{n(n+1)}{2}, \quad T_n = \sum_{\substack{1 \leq k_1, k_2 \leq n \\ k_1 + k_2 \text{ odd}}} \sqrt{k_1 k_2} \left[\frac{1}{(k_1 - k_2)^2} - \frac{1}{(k_1 + k_2)^2} \right]. \quad (28)$$

The asymptotic claim of Lemma 3.4(d.ii) is equivalent to

$$\frac{T_n}{Z_n} = \frac{\pi^2}{4} - \frac{\log n}{n} + O\left(\frac{1}{n}\right), \quad (29)$$

since $\gamma_n = \pi - (4/\pi)(T_n/Z_n) = (4/\pi)(\log n/n) + O(1/n)$.

Step 1: Diagonal-dominant decomposition. Write $k_2 = k_1 + d$ with d odd, $d \in \{\pm 1, \pm 3, \dots\}$, $k_1 \in \{1, \dots, n\}$, $k_1 + d \in \{1, \dots, n\}$. By the $k_1 \leftrightarrow k_2$ symmetry of T_n , we may take $d > 0$ and double:

$$T_n = 2 \sum_{\substack{d > 0 \\ d \text{ odd}}} \sum_{a=1}^{n-d} \sqrt{a(a+d)} \left[\frac{1}{d^2} - \frac{1}{(2a+d)^2} \right].$$

Step 2: Leading-order $\pi^2/4$. The term $1/d^2$ dominates. Using $\sqrt{a(a+d)} = a\sqrt{1+d/a} = a + d/2 - d^2/(8a) + O(d^3/a^2)$ and $\sum_{a=1}^{n-d} 1 = n - d$,

$$\sum_{a=1}^{n-d} \sqrt{a(a+d)} \cdot \frac{1}{d^2} = \frac{1}{d^2} \left[\frac{(n-d)(n-d+1)}{2} + \frac{d(n-d)}{2} + O(d \log n) \right] = \frac{n^2}{2d^2} + O\left(\frac{n}{d}\right).$$

Summing over odd d from 1 to n and using $\sum_{d \text{ odd}} 1/d^2 = \pi^2/8$ (the Catalan–Euler–style identity $\sum_{d \geq 1 \text{ odd}} 1/d^2 = (1 - 2^{-2})\zeta(2) = (3/4)(\pi^2/6) = \pi^2/8$),

$$T_n = 2 \cdot \frac{n^2}{2} \cdot \frac{\pi^2}{8} + \text{lower order} = \frac{\pi^2 n^2}{8} + \text{lower order}.$$

Dividing by $Z_n = n(n+1)/2 \sim n^2/2$ gives the leading $T_n/Z_n \rightarrow \pi^2/4$, confirming $\gamma_n \rightarrow 0$.

Step 3: Sub-leading $\log n$ (two sub-coefficients, $2 - 1 = 1$). Write $\text{defect}_n := \pi^2/4 - T_n/Z_n$, so that (29) is the statement $\text{defect}_n = \log n/n + O(1/n)$, i.e. that the $\log n/n$ coefficient of the defect is exactly 1 (and $\gamma_n = (4/\pi)\text{defect}_n$ identically by (28)). Reorganise the Step-1 double sum by a , with d odd and $d \leq n - a$, and expand $\sqrt{a(a+d)} = a + d/2 + r(a, d)$. Two pieces carry a logarithm:

- (A) *Triangle truncation of the d -sum.* The $a \cdot (1/d^2)$ piece is $2 \sum_{a=1}^{n-1} a S_2(n-a)$ with $S_2(D) = \sum_{d \text{ odd} \leq D} d^{-2} = \pi^2/8 - 1/(2D) + O(D^{-2})$ (Euler–Maclaurin on the odd- $\zeta(2)$ tail). The tail contributes $-\sum_{a=1}^{n-1} a/(n-a) = -nH_{n-1} + O(n)$, so this piece alone gives defect log-coefficient 2.

- (B) *The $d/2$ correction.* The $(d/2) \cdot (1/d^2)$ piece is $\sum_{a=1}^{n-1} S_1(n-a)$ with $S_1(D) = \sum_{d \text{ odd} \leq D} d^{-1} = \frac{1}{2} \log D + O(1)$, which is $\frac{1}{2} n \log n + O(n)$; it *adds* to T_n/Z_n and so reduces the defect with log-coefficient 1.

The remaining pieces — the $-1/(2a+d)^2$ bracket and the residual $r(a,d)$ — carry no $\log n$ at leading order: the net coefficient on the exact T_n is $2 - 1 = 1$. **[MEASURED]** All three statements are pinned independently by doubling (Richardson) estimators on the genuinely computed partial sums, none of which assumes the answer: the piece-(A) estimator converges to 2, the piece-(B) estimator to 1, and the estimator on the exact defect _{n} to 1 (`tests/test_trunk_qa_fejer_4_over_pi.py:test_subcoeff_A_triangle_truncation_is_two`, `test_subcoeff_B_sqrt_correction_is_one`, `test_net_defect_coefficient_converges_to_one`). Hence

$$\frac{T_n}{Z_n} = \frac{\pi^2}{4} - \frac{\log n}{n} + O\left(\frac{1}{n}\right),$$

which is (29). (An earlier version of this step attributed the logarithm to the $-1/(2a+d)^2$ bracket and to a boundary term in $\sqrt{a(a+d)}$, each “ $-\pi^2 n \log n/16$ ”; that bookkeeping did not compose to the displayed result and is replaced by the test-backed decomposition above, 2026-09-02.)

Step 4: Conversion to γ_n . Substituting into (28),

$$\gamma_n = \pi - \frac{4}{\pi} \left(\frac{\pi^2}{4} - \frac{\log n}{n} + O(1/n) \right) = \frac{4 \log n}{\pi n} + O\left(\frac{1}{n}\right),$$

which is the asymptotic of Lemma 3.4(d.ii).

Numerical verification. **[MEASURED]** The closed form $n\gamma_n = (4/\pi) \log n + b + O(1/n)$ with $b \approx 4.109$ has been verified to high precision against the sum-rule (28) at $n \in \{2, \dots, 1000\}$, with the doubling estimator $a_n := (2n\gamma_{2n} - n\gamma_n)/\log 2 \rightarrow 4/\pi$ reaching 1.2785 at $n = 800$ and 1.2761 at $n = 1600$ (three digits of $4/\pi$; residual shrinking $\sim 1.85 \times$ per doubling) (`tests/test_trunk_qa_fejer_test_gamma_doubling_converges_to_4_over_pi`, `test_su2_doubling_estimator_at_n800` and the slow-marked `test_su2_doubling_estimator_at_n1600`; the sum-rule itself is checked against Gaussian quadrature of the kernel moment in `tests/test_central_fejer_su2.py`).

Relation to the circle Fejér estimate. The constant $4/\pi$ is *twice* the constant of the classical estimate $\frac{1}{2\pi} \int_{\mathbb{T}^1} F_n(\theta) |\theta| d\theta \sim (2/\pi) \log n/n$ for the probability-normalised Fejér kernel on the circle (Zygmund [29] Vol. I, Ch. III; Remark 3.5, where the circle constant is computed in closed form and measured). The $\sin^2(\chi/2)$ factor in the $SU(2)$ conjugacy-class measure and the $\sqrt{2j+1}$ Plancherel weighting of Definition 3.3 together change the kernel and the constant while preserving the leading $\log n/n$ behaviour. **[OBSERVATION]** Both constants are of the form $2\text{Vol}(\text{base})/\text{Vol}(\text{group}) = 2\text{Vol}(S^0)/\text{Vol}(S^1) = 2/\pi$ and $2\text{Vol}(S^2)/\text{Vol}(SU(2)) = \text{Vol}(S^2)/\pi^2 = 4/\pi$ on the $SU(2)/U(1)$ Haar quotient — but no derivation of the factor 2 from the circle constant is given. (An earlier version of this paragraph stated the circle constant as $4/\pi$ and the $SU(2)$ constant as “preserved across this transition”; withdrawn 2026-09-02.)

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